

0064/12622545/1023

Direction For Use



B. Braun 3% Sodium Chloride Intravenous Infusion B.P.

1 NAME OF THE MEDICINAL PRODUCT

B. Braun 3% Sodium Chloride Intravenous Infusion B.P.

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

1000 ml of solution contains  
Sodium chloride 30.0 g

Electrolyte concentrations:

Sodium 513 mmol/l  
Chloride 513 mmol/l

Excipients

For the full list of excipients see section 6.1.

3 PHARMACEUTICAL FORM

Solution for infusion,  
Clear, colourless aqueous solution

Theoretical osmolality: 1026 mOsm/l  
pH: 4.5 – 7.0

4 CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of:  
• Hyponatraemia  
• Hypochloraemia  
• Hypotonic hyperhydration

4.2 Posology and method of administration

Posology

The dosage is adjusted according to the actual electrolyte requirements.

A sodium deficit is calculated according to the following formula:  
 $Na_{+required} \text{ (mmol)} = (Na_{+normal} - Na_{+actual}) \times \text{kg BW} \times 0.2$   
(The term "kg BW × 0.2" represents the extracellular volume, BW = body weight)

Adults

A daily sodium intake of 3 – 6 mmol/kg BW should not be exceeded if possible.

Paediatric population

A daily sodium intake of 3 – 5 mmol/kg BW should not be exceeded if possible.

Elderly

As for adults. However, attention should be paid to the patient's cardio-circulatory status.

Maximum infusion rate

The maximum infusion rate depends on the clinical situation.

In patients with chronic hyponatraemia the infusion rate should be sufficiently low so that the serum sodium level does not rise faster than 0.35 mmol×l<sup>-1</sup>×h<sup>-1</sup>. See also section 4.4.

Method of administration

Intravenous use  
For central venous infusion only

4.3 Contraindications

- Hypernatraemia
- Hyperchloraemia
- Isotonic hyperhydration
- Hypertonic hyperhydration

4.4 Special warnings and precautions for use

The solutions should only be administered with caution to patients with:

- Hypokalaemia
- Diseases requiring restriction of sodium intake, such as:
  - cardiac insufficiency,
  - generalized oedema
  - pulmonary oedema,
  - hypertension,
  - eclampsia,
  - severe disturbances of kidney function.

In patients with chronic hyponatraemia, a too rapid increase of the serum sodium level may lead to organic brain damage, the "osmotic demyelination syndrome". Care must be taken to keep the increase of the serum sodium level within the limit stated in section 4.2.

It is necessary to monitor the serum electrolyte levels, acid-base and water balance.

4.5 Interactions with other medicinal products and other forms of interaction

Therapy with corticosteroids or ACTH may be associated with an increased retention of sodium and water, possibly leading to oedema and hypertension.

4.6 Fertility, pregnancy and lactation

There are no data from the use of higher concentrated sodium chloride solutions for infusion in pregnant and lactating woman. In the intended indications no risks have to be expected, when volume, electrolyte and acid/base levels are carefully monitored.

Caution should be exercised in the presence of toxemia of pregnancy.

4.7 Effects on ability to drive and use machines

This medicinal product has no influence on the ability to drive and use machines.

4.8 Undesirable effects

Summary of the safety profile

Undesirable effects are mainly caused by inadequate dosages and/or infusion rates. Under conditions of proper administration they will occur only rarely, i.e. in less than 1/1000 of treated patients.

Listing of adverse reactions

Metabolism and nutrition disorders

Sodium chloride administration may lead to an increase of blood sodium and chloride levels (hypernatraemia, hyperchloraemia). Symptoms of severe hypernatraemia see section 4.9.

Rapid infusion may lead to acute volume overload.

Gastrointestinal disorders

Diarrhoea due to inadequately high infusion rates

Renal and urinary disorders

Osmotically induced polyuria (osmotic diuresis) due to inadequately high infusion rates

Approval for Printing

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4.9 Overdose

Symptoms

Overdose can lead to hyperhydration, hypernatraemia, hyperchloraemia, hyperosmolarity and disturbances of the acid-base balance, in particular acidosis.

Clinical symptoms of hypernatraemia:

Thirst, motor unrest, hyperexcitability of muscles, hypertension, vertigo, vomiting, somnolence, dryness of skin and mucosae, fever, oedema, congestive heart failure, hyperosmolar coma.

Treatment

Immediate stop of infusion, administration of diuretics with continuous monitoring of the electrolyte balance, correction of electrolyte disturbances, counteraction of acidosis.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Solutions affecting the electrolyte balance -Electrolytes  
ATC code:B05B B01

Mechanism of action

3% w/v Sodium Chloride Intravenous Infusion and Sodium Chloride Intravenous Infusion 500 mmol/l are hypertonic solutions of the most abundant ions in the body fluids, namely sodium and chloride.

Sodium is the primary cation of the extracellular space and together with various anions, regulates the size of this. Sodium and potassium are the major mediators of bioelectric processes within the body.

The sodium content and the fluid metabolism of the body are closely coupled to each other. Each deviation of the plasma sodium concentration from the physiological one simultaneously affects the fluid status of the body.

An increase in the sodium content of the body results also means reduction of the body's free water content independent of the serum osmolality.

Pharmacodynamic effect

Sodium Chloride 30 mg/ml and Sodium Chloride 500 mmol/l are mainly used for rapid sodium and chloride substitution in situations of serious depletion of those electrolytes.

5.2 Pharmacokinetic properties

Absorption

As the solutions are administered directly into the circulation, their bio-availability is 100 per cent.

Further pharmacokinetic properties

The total sodium content of the body is ca. 80 mmol/kg of which ca. 97 % is extracellular and ca. 3 % intracellular. The daily turnover is ca. 100 - 180 mmol (corresponding to 1.5 - 2.5 mmol/kg body weight).

The kidneys are the major regulator of the sodium and water balances. In co-operation with the hormonal control mechanisms (renin-angiotensin-aldosterone system, antidiuretic hormone) and the hypothetical natriuretic hormone they are primarily responsible for keeping the volume of the extracellular space constant and regulating its fluid composition.

Chloride is exchanged for hydrogen carbonate in the tubule system and is, thus, involved in the regulation of the acid-base balance.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Water for Injections

6.2 Incompatibilities

The compatibility of any additive or drug to be admixed must be checked before use. See also section 6.6.

6.3 Shelf life

3 years

6.4 Special precautions for storage

Do not store above 30 °C.

For storage conditions after admixture of additives, see section 6.3.

6.5 Nature and contents of container

Bottles of colourless low-density polyethylene, contents: 500 ml

6.6 Special precautions for disposal and other handling

Use only if the solution is clear and without visible particles. The solution should not be administered if it is not clear or if the container or its closure show visible signs of damage.

For single use only. Discard unused contents. Do not reconnect partially used containers.

Administer immediately following the insertion of the infusion set.

The solution should be administered with sterile equipment using an aseptic technique. The equipment should be primed with the solution in order to prevent air entering the system.

As with all parenteral solutions, before adding medications, compatibility of these additives with the solution in the container must be assessed. Before introducing an additive verify it is soluble and stable in water.

Incompatibility of an additive medication with the sodium chloride solutions is to be judged by checking for colour change and/or precipitate, insoluble complexes or crystals that may appear.

Thorough and careful aseptic mixing of any additive is mandatory. Solutions containing additives should be used immediately after preparation, unless preparation has taken place in controlled and validated aseptic conditions.

Those additives known to be incompatible should not be used.

The Instructions for Use of the medication to be added must be consulted.

When a compatible medication is added to this formulation, the solution must be administered immediately, unless dilution has taken place in controlled and validated aseptic conditions.

7. DATE OF REVISION OF THE TEXT

January 2023

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Product registration holder and  
manufactured by:  
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